

Design and Implementation of a Three-Band Analog Audio Preamplifier and Equalizer

Abstract—This paper presents the redesign and simulation of a three-band analog audio preamplifier and equalizer in LTspice. The circuit consists of an input buffer, separate bass, midrange, and treble filter stages, adjustable band-level controls, a summing amplifier, a master volume control, and an equalizer bypass switch. The filters were readjusted from the original reference design, targeting corner frequencies near 200 Hz for bass, 300 Hz–4 kHz for midrange, and 6 kHz for treble. AC simulations were used to check each stage's individual gain and their cutoff behavior.

A unity-gain phase-inversion stage was added to the treble path to reduce destructive interference at the summing node. In the absence of the unity-gain inversion stage, the treble and midrange responses experience partial cancellation. The final circuit gives independent control over the three bands, a bypass switch, and an adjustable output level. LTspice simulations were used to confirm the frequency response and signal levels of both the individual stages as well as the complete circuit.

Index Terms—analog audio, audio preamplifier, three-band equalizer, bass, midrange, treble, active filter, TL072, Sallen-Key, multiple-feedback filter, LTspice

I. INTRODUCTION

Audio equalization matters because it lets you reshape the frequency content of a signal to obtain a specific tonal character. The lower frequencies carry the bass, the midrange gives the audio its body and clarity, and the higher frequencies carry treble and detail. A multi-band equalizer controls these regions independently and changes the overall frequency response significantly.

Analog equalizers do this with resistors, capacitors, and op-amps. Active topologies such as Sallen-Key and multiple-feedback filters work well here since they provide controlled cutoff frequencies and gain without needing inductors. Op-amps also handle buffering, amplification, and summing, allowing the separate filter stages to be combined into one system.

The project started from an existing three-band audio preamplifier reference design available on GitHub [1]. The work presented here redesigns the filter characteristics and extends the signal-processing stages to meet the required specifications. The work presented here redesigns the filter characteristics and extends the signal-processing stages to meet the required specifications. The redesign was carried out in LTspice, allowing the modified circuit to be evaluated using frequency-domain simulations before settling on a final configuration.

The redesigned system has an input buffer followed by independent bass, midrange, and treble filter stages. Each band has an adjustable level control, and the three signals then

combine through an inverting summing amplifier. A master volume control is provided at the output, along with an equalizer bypass that switches between the processed signal and the buffered input. A unity-gain phase-inversion stage was added in the treble path to reduce partial cancellation between the treble and midrange components at the summing stage. Both the individual stages and the full circuit were checked using LTspice frequency-response simulations.

II. BLOCK DIAGRAM

The overall signal flow of the redesigned audio preamplifier and equalizer is shown in Fig. 1. The input signal is first buffered and then divided into three frequency-selective paths for bass, midrange, and treble processing. Each band has an independent level control. The controlled signals are combined using a summing amplifier, followed by the master volume control. An equalizer bypass path is also provided to allow the buffered input signal to be selected directly.

The buffered signal is denoted by $in-L$. The equalizer output is denoted by L_OUT . The selector chooses one of these signals and routes the selected signal, denoted by COM , to the master-volume stage.

III. THEORY AND DESIGN

A. Input Buffer

The input stage is implemented as a TL072 voltage follower. The input signal is connected to the non-inverting terminal, while the output is directly fed back to the inverting terminal. Hence,

$$A_v \approx 1. \quad (1)$$

The voltage follower provides high input impedance and low output impedance, minimizing loading of the source and isolating it from the frequency-selective network. Its output, $in-L$, is also used as the bypass signal.

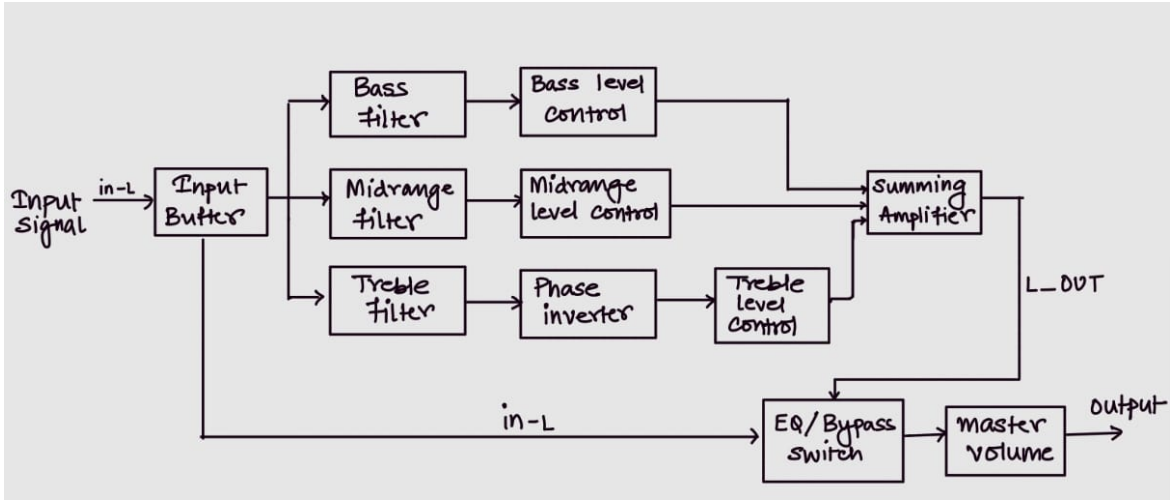


Fig. 1. Block diagram of the redesigned three-band analog audio preamplifier and equalizer.

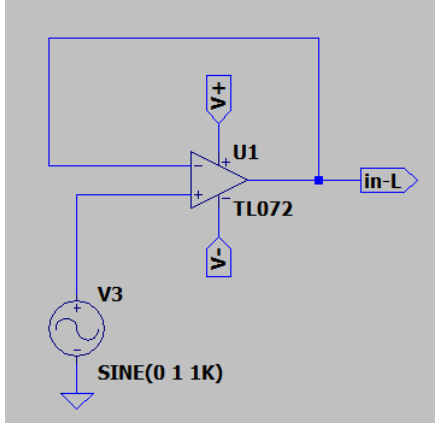


Fig. 2. Input voltage-follower stage.

B. Bass Control Stage

The bass channel uses a second-order multiple-feedback low-pass filter. The final component values are $R_1 = 100\text{ k}\Omega$, $R_2 = 39\text{ k}\Omega$, $R_3 = 470\text{ k}\Omega$, $C_1 = 1.6\text{ nF}$, and $C_2 = 22\text{ nF}$. The transfer function for the implemented topology is

$$H(s) = \frac{-R_3/R_1}{1 + sC_1 \left(R_2 + R_3 + \frac{R_2 R_3}{R_1} \right) + s^2 C_1 C_2 R_2 R_3}. \quad (2)$$

The low-frequency voltage gain is therefore approximately

$$A_{v,B} = -\frac{R_3}{R_1} = -\frac{470}{100} = -4.7, \quad (3)$$

corresponding to a magnitude of approximately 13.44 dB. The capacitor impedance is given by

$$Z_C = \frac{1}{sC}. \quad (4)$$

For sinusoidal steady-state operation, $s = j2\pi f$, giving

$$X_C = \frac{1}{2\pi fC}. \quad (5)$$

The selected capacitor values shift the bass transition toward the target frequency of approximately 200 Hz. LTspice AC analysis produced a measured transition near 203 Hz.

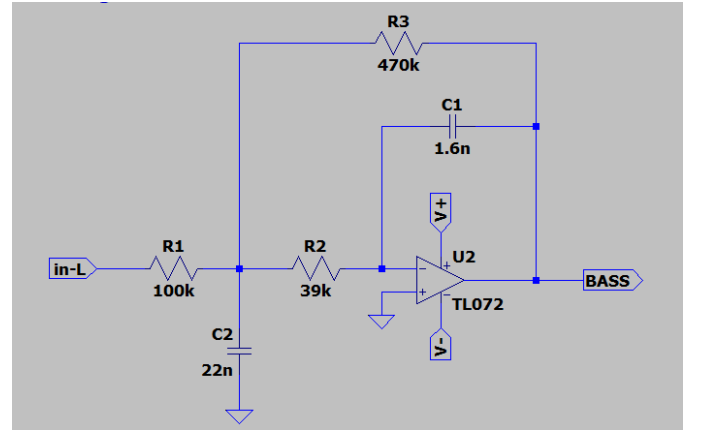


Fig. 3. Second-order multiple-feedback low-pass filter used for the bass channel.

C. Midrange Control Stage

The midrange channel is formed by cascading a second-order high-pass filter and a second-order low-pass filter. For the equal-component Sallen-Key sections, the characteristic frequency is

$$f_0 = \frac{1}{2\pi RC}. \quad (6)$$

The redesigned high-pass stage uses $R_4 = R_7 = 780\text{ k}\Omega$ and $C_3 = C_4 = 680\text{ pF}$. Thus,

$$f_{0,HP} = \frac{1}{2\pi(780 \times 10^3)(680 \times 10^{-12})} \approx 300.3\text{ Hz}. \quad (7)$$

Its non-inverting gain is determined by R_5 and R_6 :

$$A_{v,HP} = 1 + \frac{R_6}{R_5} = 1 + \frac{56}{100} = 1.56, \quad (8)$$

corresponding to approximately 3.86 dB. LTspice produced an individual-stage transition near 306 Hz.

The redesigned low-pass stage uses $R_{10} = R_{11} = 270 \text{ k}\Omega$ and $C_5 = C_6 = 150 \text{ pF}$. Hence,

$$f_{0,LP} = \frac{1}{2\pi(270 \times 10^3)(150 \times 10^{-12})} \approx 3.93 \text{ kHz}. \quad (9)$$

Its non-inverting gain is determined by R_{12} and R_{13} :

$$A_{v,LP} = 1 + \frac{R_{13}}{R_{12}} = 1 + \frac{56}{100} = 1.56, \quad (10)$$

corresponding to approximately 3.86 dB. LTspice produced an individual upper transition near 3.855 kHz. The cascaded midrange response produced lower and upper -3 dB frequencies of approximately 302 Hz and 3.875 kHz, respectively.

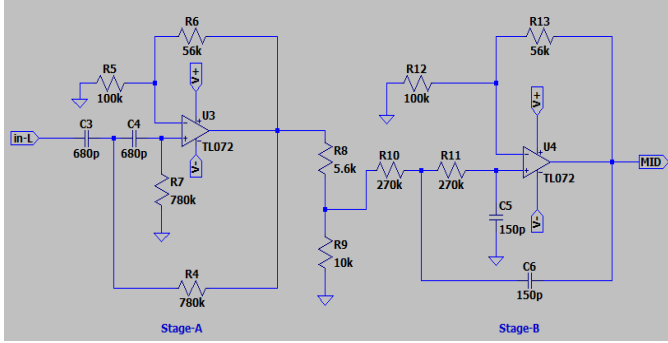


Fig. 4. Redesigned cascaded high-pass and low-pass filter stages used for the midrange channel.

D. Treble Control Stage

The treble channel uses a second-order Sallen-Key high-pass filter. The final component values are $R_{14} = R_{17} = 270 \text{ k}\Omega$, $C_7 = C_8 = 90 \text{ pF}$, $R_{15} = 100 \text{ k}\Omega$, and $R_{16} = 56 \text{ k}\Omega$.

For the equal-component approximation, the characteristic frequency is

$$f_{0,T} = \frac{1}{2\pi RC}. \quad (11)$$

Using $R = 270 \text{ k}\Omega$ and $C = 90 \text{ pF}$ gives

$$f_{0,T} = \frac{1}{2\pi(270 \times 10^3)(90 \times 10^{-12})} \approx 6.55 \text{ kHz}. \quad (12)$$

The non-inverting stage gain is determined by R_{15} and R_{16} :

$$A_{v,T} = 1 + \frac{R_{16}}{R_{15}} = 1 + \frac{56}{100} = 1.56, \quad (13)$$

corresponding to approximately 3.86 dB. LTspice produced an individual-stage transition near 6.016 kHz.

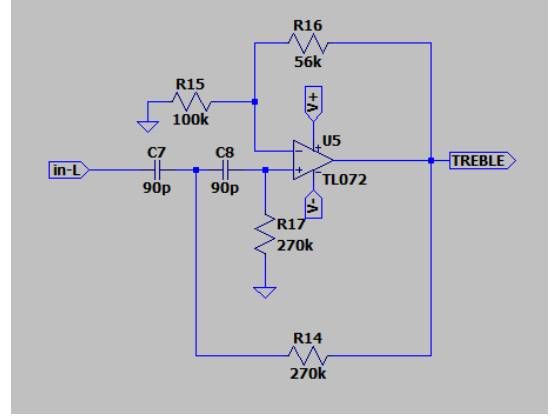


Fig. 5. Second-order Sallen-Key high-pass filter used for the treble channel.

E. Band-Control Potentiometers

Each frequency band is provided with an independent level control using a parameterized 10-k Ω hierarchical potentiometer. The three potentiometers independently adjust the amount of bass, midrange, and treble signal passed to the summing stage. Thus, the relative contribution of each frequency band to the final output can be adjusted without changing the filter characteristics.

The potentiometer is modeled in LTspice using two variable resistances:

$$R_a = R \left(1 - \frac{\text{slider}}{100} \right) + 10^{-6}, \quad (14)$$

$$R_b = R \left(\frac{\text{slider}}{100} \right) + 10^{-6}. \quad (15)$$

A small offset is included to avoid an exact zero-resistance condition at the potentiometer endpoints in the LTspice model.

For an unloaded voltage-divider configuration, the wiper voltage is

$$V_w = V_{in} \frac{R_b}{R_a + R_b}. \quad (16)$$

At a 50% slider position,

$$R_a = R_b = 5 \text{ k}\Omega, \quad \frac{V_w}{V_{in}} = 0.5, \quad (17)$$

giving an ideal attenuation of

$$20 \log_{10}(0.5) = -6.02 \text{ dB}. \quad (18)$$

In the complete circuit, the following stage loads the potentiometer wiper, so the actual attenuation can differ slightly from the ideal unloaded value. The potentiometers therefore provide independent control of the bass, midrange, and treble levels before the signals are combined by the summing amplifier.

F. Inverting Summing Amplifier

The controlled bass, midrange, and treble signals are combined using a TL072 configured as an inverting summing amplifier. The three band signals are applied to the inverting input through equal input resistors, while the non-inverting input is grounded. The input resistors are $R_1 = R_{21} = R_{22} = 56\text{ k}\Omega$, and the feedback resistor is $R_{20} = 560\text{ k}\Omega$.

For an ideal op-amp, the output voltage is given by

$$V_{U6} = -R_{20} \left(\frac{V_{SW_CON}}{R_1} + \frac{V_{MID_CON}}{R_{21}} + \frac{V_{TREBLE_CON}}{R_{22}} \right). \quad (19)$$

Since the three input resistors are equal,

$$V_{U6} = -10 (V_{SW_CON} + V_{MID_CON} + V_{TREBLE_CON}). \quad (20)$$

Thus, each individual input has a nominal magnitude gain of 10, corresponding to 20 dB, when considered independently. The summing stage therefore performs both signal mixing and amplification, with the contribution of each frequency band determined by its respective level-control potentiometer.

During complete-system simulation, the bass, midrange, and treble signals had different phase relationships, which caused partial cancellation when they were combined at the summing stage. To reduce this effect, a unity-gain inverting stage was added to the treble path before the summing amplifier. The phase-inversion stage uses equal input and feedback resistors of $10\text{ k}\Omega$, giving

$$A_{v,U7} = -\frac{R_f}{R_{in}} = -1. \quad (21)$$

The stage therefore preserves the magnitude of the treble signal while reversing its polarity. This changes its phase by 180° relative to the signal entering the inverter and reduces the observed cancellation between the treble and midrange contributions at the summing node.

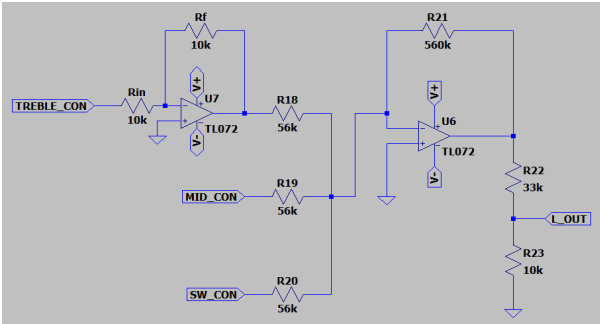


Fig. 6. TL072 inverting summing amplifier and associated output-stage circuitry.

The output of the summing amplifier is then passed through a resistive voltage divider consisting of $R_{24} = 33\text{ k}\Omega$ and $R_{25} = 10\text{ k}\Omega$. The output voltage is therefore

$$V_{L_OUT} = V_{U6OUT} \frac{R_{25}}{R_{24} + R_{25}}, \quad (22)$$

which gives

$$\frac{V_{L_OUT}}{V_{U6OUT}} = \frac{10}{33 + 10} \approx 0.2326. \quad (23)$$

Thus, the divider attenuates the summing-amplifier output by approximately -12.67 dB before it is sent to the output selector and master volume control.

G. EQ/Bypass Selector

An EQ/bypass selector is placed after the equalizer to allow either the processed signal or the original buffered signal to be selected. The selector chooses between the equalized output L_OUT and the buffered input signal $in-L$ and passes the selected signal to the master-volume potentiometer.

In EQ mode,

$$COM = L_OUT, \quad (24)$$

while in bypass mode,

$$COM = in-L. \quad (25)$$

The EQ/bypass selection is implemented in LTspice using two complementary voltage-controlled switches. When the EQ path is selected, L_OUT is connected to COM , while in bypass mode, $in-L$ is connected to COM . The switches are controlled with opposite control signals so that only one path is active at a time.

The switch model used is

$$\text{.model SWMOD SW (Ron=1 Roff=1Meg Vt=2.5 Vh=0.1)}. \quad (26)$$

The control voltage is used only to select the switch position during simulation and does not affect the audio signal itself.

H. Master-Volume Control

The selected signal COM is applied to X4, a parameterized $100\text{-k}\Omega$ potentiometer. The wiper of the potentiometer provides the final output signal, $MASTER_OUT$. The potentiometer controls the overall output level without changing the frequency characteristics of the equalizer.

For an unloaded voltage-divider configuration, the output voltage at a slider fraction α is approximately

$$\frac{V_{MASTER_OUT}}{V_{COM}} \approx \alpha. \quad (27)$$

At 100% slider position, the expected attenuation is approximately 0 dB . At 50% slider position,

$$20 \log_{10}(0.5) = -6.02\text{ dB}. \quad (28)$$

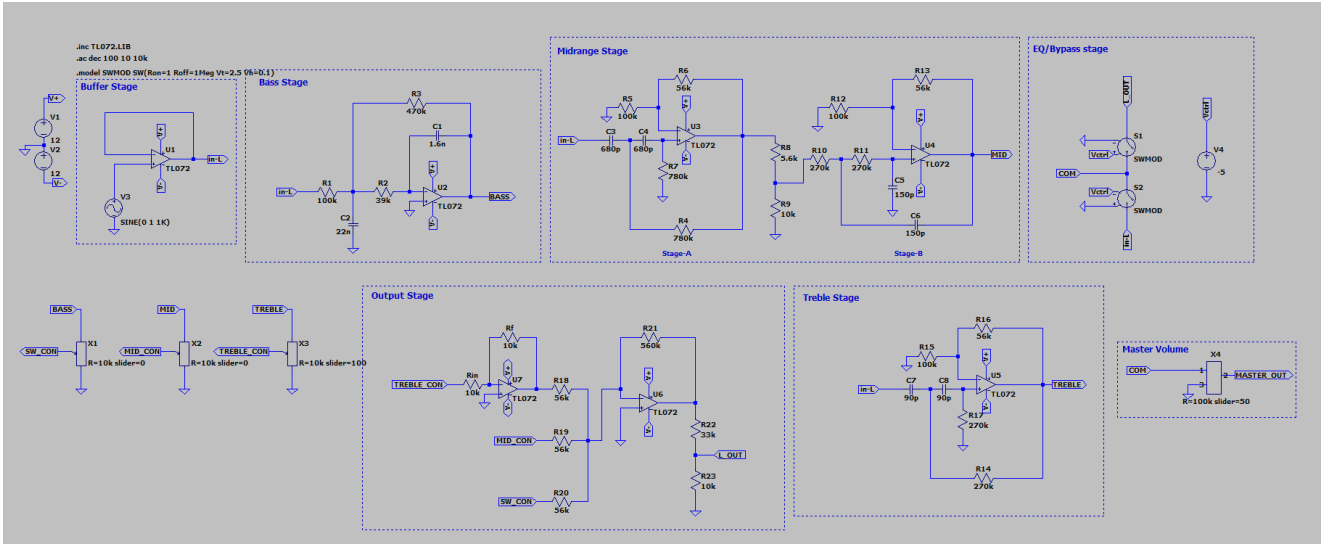


Fig. 7. Complete redesigned three-band analog audio preamplifier and equalizer.

TABLE I
PRINCIPAL REDESIGN AND LTSPICE VERIFICATION RESULTS

Stage	Target	Measured	Error
Bass	200 Hz	203 Hz	1.5%
Mid HP	300 Hz	306 Hz	2.0%
Mid LP	4 kHz	3.855 kHz	3.6%
Mid cascade, lower	—	302 Hz	—
Mid cascade, upper	—	3.875 kHz	—
Treble	6 kHz	6.016 kHz	0.27%

IV. FINAL CIRCUIT

The final schematic as shown in Fig.7 combines the re-designed filters, custom potentiometers, summing stage, attenuation network, EQ/bypass selector, and master-volume control.

The final signal flow is *Input* → *Buffer* → {*Bass*, *Mid*, *Treble*} → *Pots* → *Summing* → *Attenuation* → *Selector* → *Master Volume* → *Output*.

The bypass branch is taken from *in-L*, so the equalizer can be removed from the signal path without removing the input buffer or master-volume function.

V. SIMULATION RESULTS

The redesigned audio preamplifier and equalizer were simulated in LTspice to verify the frequency response and signal levels of the individual stages and the complete circuit. The simulated results were compared with the expected values obtained from the circuit design. Table I summarizes the principal redesign targets and measured LTspice values.

A. Input Buffer Verification

AC analysis gave approximately 0-dB voltage gain for the voltage follower, confirming unity transfer of the buffered input.

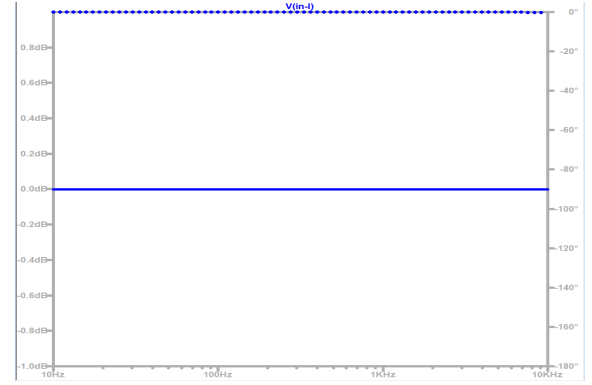


Fig. 8. AC response of the input voltage follower.

B. Bass Verification

The redesigned bass filter produced a transition near 203 Hz, confirming the shift toward the intended 200-Hz region.

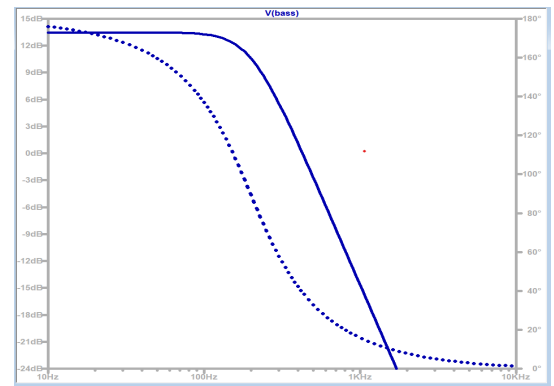


Fig. 9. Simulated frequency response of the bass filter.

C. Midrange Verification

The redesigned high-pass and low-pass stages produced transitions near 306 Hz and 3.855 kHz, respectively. Their cascade produced approximately 302 Hz and 3.875 kHz as the lower and upper -3-dB frequencies.

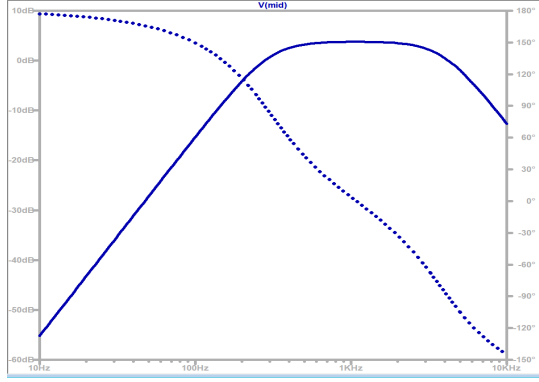


Fig. 10. Simulated frequency response of the Midrange filter.

D. Treble Verification

The treble filter produced a transition near 6.016 kHz. The additional unity-gain inverting stage reduced destructive phase interaction observed during complete-system summation.

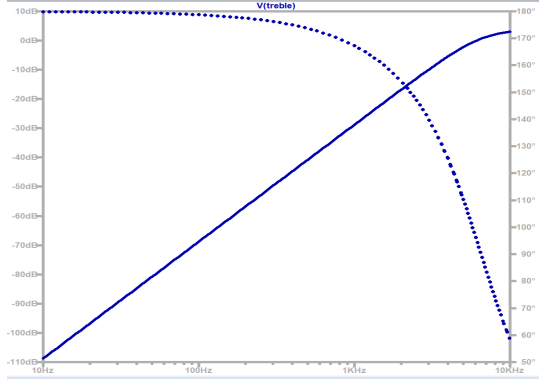


Fig. 11. Simulated frequency response of the Treble filter.

E. Potentiometer Verification

Each frequency band is provided with an independent level control using a parameterized 10-k Ω potentiometer. The potentiometer acts as a variable voltage divider and controls the amplitude of the corresponding band before it is applied to the summing amplifier. Therefore, the potentiometer changes the relative contribution of the bass, midrange, or treble signal without changing the cutoff frequency of the associated filter.

The potentiometer is modeled using two resistances, R_a and R_b , whose values vary with the slider position. They are defined as

$$R_a = R \left(1 - \frac{\text{slider}}{100} \right) + 10^{-6}, \quad (29)$$

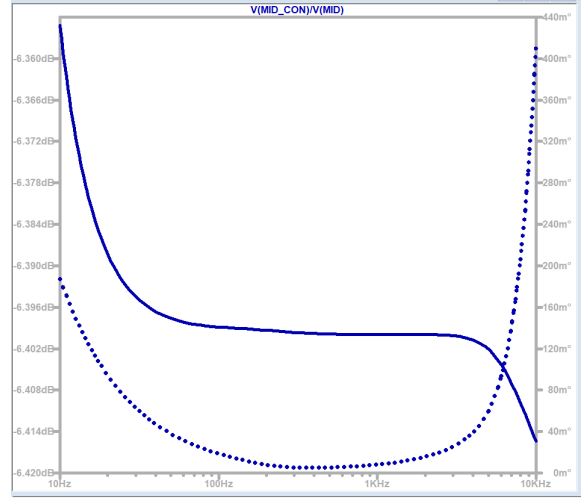


Fig. 12. Simulated attenuation of the midrange level-control potentiometer at a 50% slider position

$$R_b = R \left(\frac{\text{slider}}{100} \right) + 10^{-6}. \quad (30)$$

For an ideal unloaded voltage-divider configuration, the wiper voltage is

$$V_w = V_{in} \frac{R_b}{R_a + R_b}. \quad (31)$$

Thus, the slider position determines the fraction of the input signal appearing at the wiper. At 0% slider position, the wiper approaches the lower end of the potentiometer and the output approaches zero. At 100%, the wiper approaches the input end and the output approaches the full input voltage. At the 50% position,

$$R_a = R_b = 5 \text{ k}\Omega, \quad (32)$$

giving

$$\frac{V_w}{V_{in}} = \frac{5}{5 + 5} = 0.5. \quad (33)$$

The corresponding ideal attenuation is

$$20 \log_{10}(0.5) = -6.02 \text{ dB}. \quad (34)$$

In the complete circuit, the wiper is connected to the input of the summing amplifier and is therefore not perfectly unloaded. The effective loading changes the divider ratio slightly from its ideal value. For a 50% slider position, the simulated attenuation was approximately -6.4 dB, which is close to the theoretical value.

F. Summing and Output Verification

The outputs of the bass, midrange, and treble level-control potentiometers are combined using a TL072 configured as an inverting summing amplifier. Each band is applied to the

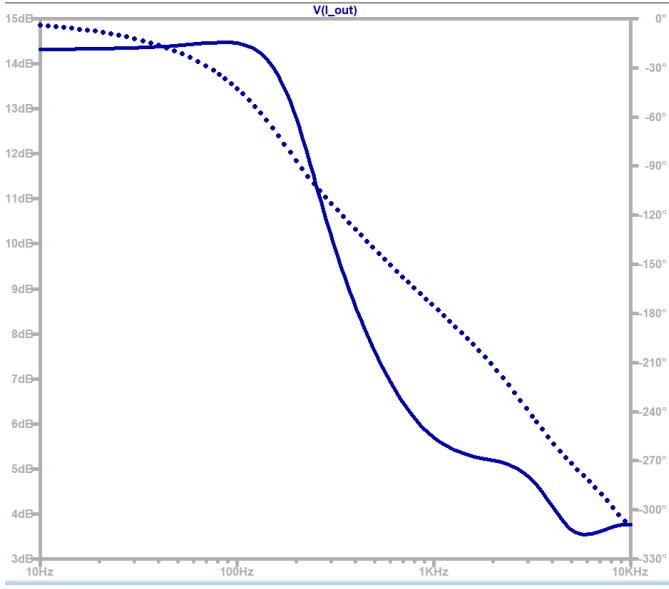


Fig. 13. Simulated output response after the summing amplifier and output attenuation stage

inverting input through an individual $56\text{ k}\Omega$ resistor, while the feedback resistor is $560\text{ k}\Omega$.

For an ideal inverting summing amplifier, the output is

$$V_{U6} = -R_{20} \left(\frac{V_{SW_CON}}{R_1} + \frac{V_{MID_CON}}{R_{21}} + \frac{V_{TREBLE_CON}}{R_{22}} \right) \quad (35)$$

Since all input resistors are equal,

$$V_{U6} = -10 (V_{SW_CON} + V_{MID_CON} + V_{TREBLE_CON}). \quad (36)$$

Thus, each isolated input has a nominal gain magnitude of

$$|A_v| = \frac{560}{56} = 10, \quad (37)$$

or approximately 20 dB. The summing stage therefore combines the three independently controlled frequency bands while providing additional amplification.

The summed output is then attenuated by the $33\text{ k}\Omega/10\text{ k}\Omega$ resistive divider. Its voltage ratio is

$$\frac{V_{L_OUT}}{V_{U6OUT}} = \frac{10}{33 + 10} \approx 0.2326, \quad (38)$$

corresponding to an attenuation of approximately -12.67 dB .r provides approximately -12.67 dB attenuation.

G. EQ/Bypass Verification

In EQ mode, LTspice gave

$$\frac{V(COM)}{V(L_OUT)} \approx 0\text{ dB}, \quad (39)$$

confirming that the equalized signal is selected. In bypass mode,

$$\frac{V(COM)}{V(in-L)} \approx 0\text{ dB}, \quad (40)$$

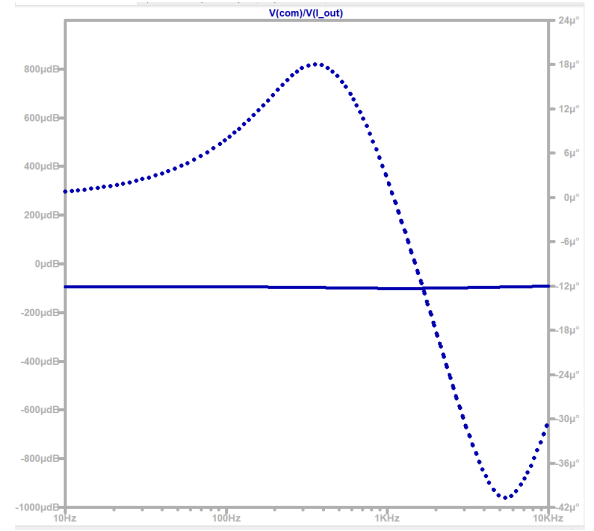


Fig. 14. EQ-mode selector verification showing the transfer from L_OUT to COM .

confirming that the original buffered signal is selected.

H. Master-Volume Verification

The master-volume potentiometer was verified at 50% and 100% slider positions. At 50%, the measured attenuation was approximately -6.02 dB , while at 100%, the attenuation approached 0 dB . These results agree with the expected behavior of the voltage-divider configuration.

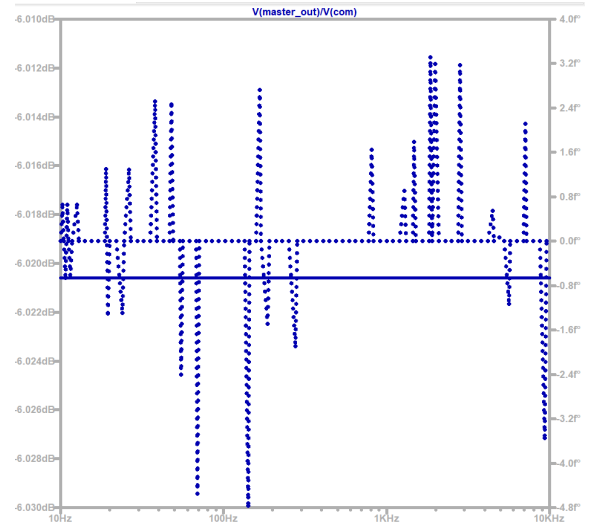


Fig. 15. Master-volume response at 50% and 100% slider positions.

I. Complete-System Verification

The complete redesigned circuit was simulated to verify the overall frequency response from the buffered input to the final output. The result includes the three-band equalizer, band-level controls, summing stage, output attenuation, EQ/bypass selector, and master-volume control.

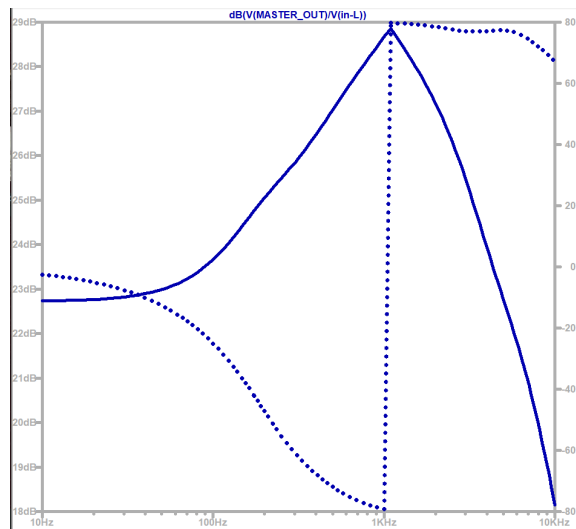


Fig. 16. Complete-system frequency response of the redesigned audio preamplifier and equalizer.

VI. DISCUSSION

The redesign shows that the frequency response of the individual filter stages can be shifted by changing selected resistor and capacitor values while keeping the same basic filter topologies. In the bass stage, the capacitor values were modified to move the low-pass transition toward 200 Hz. For the midrange section, the resistor values were adjusted to place the lower and upper transitions near 300 Hz and 4 kHz, respectively. The treble-stage resistor values were also modified to obtain a transition close to 6 kHz.

The custom hierarchical potentiometers provide independent control of the bass, midrange, and treble levels, as well as the overall output level. Their simulation also shows that the actual attenuation of a potentiometer can differ slightly from the ideal voltage-divider value because the wiper is loaded by the following circuit.

An important system-level observation was the effect of phase differences when the three filtered signals are combined. Since the individual band signals can have different amplitudes and phases, partial cancellation can occur at the summing amplifier. A unity-gain inverting stage was therefore added to the treble path after this effect was observed in simulation. This reverses the treble signal polarity without changing its magnitude and reduces the destructive interaction at the summing stage.

The EQ/bypass selector adds a simple way to compare the processed and unprocessed signals. In EQ mode, the equalized signal is selected, while in bypass mode, the buffered input is selected. In both cases, the selected signal is passed through the same master-volume control, allowing the output level to be adjusted independently of the EQ/bypass selection.

VII. CONCLUSION

A redesigned three-band analog audio preamplifier and equalizer was implemented and evaluated using LTspice. The final system consists of an input voltage follower, separate

bass, midrange, and treble filter paths, independent band-level controls, an inverting summing amplifier, an output attenuation network, an EQ/bypass selector, and a master-volume control.

The simulations produced transition frequencies of approximately 203 Hz for the bass band, 302 Hz–3.875 kHz for the midrange band, and 6.016 kHz for the treble band. The custom potentiometer models were also verified to provide variable attenuation, while the master-volume stage produced approximately 0 dB at 100% slider position and -6.02 dB at 50%. The EQ/bypass selector successfully allowed either the equalized signal or the buffered input signal to be selected. In addition, the unity-gain phase-inversion stage in the treble path helped reduce the partial cancellation observed when the three bands were combined.

Overall, the project demonstrates the practical design and simulation of analog audio signal-processing stages, including active filtering, op-amp feedback, variable attenuation, signal summation, buffering, and phase management. The final circuit provides independent three-band equalization together with convenient EQ/bypass and master-volume controls, without relying on digital signal processing.

ACKNOWLEDGMENT

The initial implementation of this project was developed as part of an earlier group project with Shreya Arya Jaiswal and Rishabh Kothari, who happen to be my classmates. Their contributions to the initial implementation are gratefully acknowledged. The present work independently redesigns and extends that implementation, including modifications to the filter characteristics, signal-processing stages, and LTspice simulations.

REFERENCES

- [1] thiasotiks, "Audio-Preamplifier-With-3-band-Equalizer," GitHub repository. [Online]. Available: <https://github.com/thiasotiks/Audio-Preamplifier-With-3-band-Equalizer>
- [2] Texas Instruments, "TL072 Data Sheet, Product Information and Support." [Online]. Available: <https://www.ti.com/product/TL072>
- [3] J. Karki, "Active Low-Pass Filter Design," Texas Instruments, Application Report SLOA049D, 2023. [Online]. Available: <https://www.ti.com/lit/an/sloa049d/sloa049d.pdf>
- [4] A. S. Sedra and K. C. Smith, *Microelectronic Circuits*, 7th ed. New York, NY, USA: Oxford University Press, 2014.